CAN BIDIRECTIONAL LANGUAGE MODEL MITIGATE REVERSAL CURSE?

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ABSTRACT

Large Language Models(LLMs) like GPT-4, despite their impressive capabilities in understanding and generating text, are struggling with bidirectional relationships, which is the so-called Reversal Curse. A notable example is the model's ability to know that Tom Cruise's mother exists, but failing to identify her most famous son. This discrepancy suggests that models trained with autoregressive, unidirectional transformers may struggle to infer relationships in the reverse direction, even when they understand the forward relationship (A $\rightarrow$ B). While GPT-like models focus on unidirectional learning, some bidirectional models like BERT, may alleviate some of these challenges by considering both the left and right context of a given token simultaneously. This paper proposes to investigate whether bidirectional models like BERT and BART, based on the study of Berglund et al. (2023), can better handle tasks requiring the reversal of relationships, and whether it can outperform autoregressive models in such scenarios.

1 INTRODUCTION

The reversal curse, observed by Berglund et al. (2023), has drawn much attention. This phenomenon is a significant challenge in the field of natural language processing (NLP), where models trained to infer relationships in one direction (e.g., from a person to their description) struggle when asked to reverse these relationships (e.g., from a description to the person). This issue has been observed in various state-of-the-art language models, including GPT-4, which despite its impressive capabilities, often fails to predict the reverse relationships. For instance, GPT-4 can easily get the right answer about the name of a celebrity's parent. However, when tasked with identifying even the most famous child based on the name of celebrity's parent, GPT-4's performance is far from ideal. This suggests that the model, while adept at learning structural patterns, does not necessarily learn the underlying real-world associations between entities, particularly when the task requires switching the direction of reasoning. One of the primary reasons for this issue may lie in the unidirectional nature of models like GPT, which are trained to generate text in a left-to-right fashion, and thus may not be optimized for reversing learned relationships. In contrast, bidirectional models like BERT, which process context in both directions, could offer a potential solution by enabling more effective inference of reversed relationships. This paper explores the hypothesis that bidirectional models, with their ability to analyze context from both ends of a relationship, may be more capable of handling tasks that involve reversing entity relationships. To test this hypothesis, I use BERT family models to replicate the reversal curse experiment based on the research of Berglund et al. (2023), comparing their performance in predicting relationships with GPT-4, specifically in a task involving parent-child relationships. We further investigate how adjustments to training methods, including fine-tuning with GPT-4-generated data and few-shot learning, can help enhance the performance of these models in tasks that involve complex relational reasoning. Through these experiments, we aim to assess whether bidirectional models such as BERT can mitigate the reversal curse and provide more accurate predictions when tasked with reversing known relationships.

2 RELATED WORK

2.1 THE REVERSAL CURSE OF LLMS

Berglund et al. (2023) verified that fine-tuning an large language models(LLMs) on a set of relations formatted as "A is B" does not guarantee that the model will generalize to "B is A". In fact, when asked to reverse the direction of a relation, the model's output is no more likely to be correct than a random guess. This observation led to the formalization of the reversal curse, which underscores the challenge LLMs face when encountering tasks that involve reversing relationships.

Other research has sought to explore and explain this phenomenon. Grosse et al. (2023) has examined the influence of training examples on LLM outputs. Their experiments showed that training examples presented in the "A precedes B" order have a much stronger influence on the model's output than those with the reverse order. If a pretrained model has not seen relationships in both directions during training, it is unlikely to generalize to both directions, further supporting the reversal curse hypothesis. This finding suggests that LLMs, when trained predominantly on one direction of a relationship, fail to adequately handle the reverse direction.

There have been attempts to mitigate the reversal curse through various strategies, such as augmenting training data with reverse examples (Yu et al. (2023)) or employing knowledge editing techniques (Ma et al. (2023)).

Additionally, factors influencing the reversal curse during inference have also been explored. Recent works (Wang et al. (2022), Merullo et al. (2023)) suggest that mechanistic interpretability might help us better understand how models process reversibility and why they fail in this regard. While knowledge editing strategies provide a practical approach to mitigate some of these issues, more fundamental research is needed to develop an effective framework that addresses the root causes of the reversal curse.

In comparison to earlier works that focused primarily on unidirectional model editing (Meng et al. (2022)), Lv et al. (2024) introduces a novel perspective by targeting bidirectional editing and evaluation, which is the first attempt to formally quantify reversibility, build a benchmark for evaluating reversibility in model editing, and propose effective methods for mitigating the reversal curse in LLMs.

2.2 TEXT GENERATION VIA BERT FAMILY

BERT, once put forward by Devlin (2018), revolutionized the field of Natural Language Processing by(NLP) by employing the Transformer architecture and large-scale pre-training. It is an encoder-only model that incorporates a multi-layer bidirectional Transformer encoder (Vaswani (2017)), whose bidirectional nature and the parallelizable masked language model (MLM) objective allow it to learn contextual word representations, helping it capture rich semantic information. Therefore, BERT has been widely used in natural language understanding (NLU) tasks such as question answering (QA) and named entity recognition (NER), where understanding context from both directions is critical. The success of BERT has led to the creation of the BERT Family, including models like RoBERTa (Liu (2019)), ELECTRA (Clark (2020)), DeBERTa (He et al. (2020)), and XLM-R (Conneau (2019)), which further optimized the pre-training methods and improved performance across various NLU tasks.

While BERT and its variants have excelled in NLU, they have primarily been used for extracting contextual features, and not for generating text (Zhu et al. (2020)). However, recent research has explored their potential in non-autoregressive text generation. Early works by Dong et al. (2019)) and (Wang & Cho (2019)) suggested that BERT could be adapted to generate coherent text, provided the model is appropriately modified. Recent works have proposed non-autoregressive generation approaches. For instance, Yang et al. (2020) has demonstrated that BERT can be adapted for generation tasks using masked language models (MLMs) and denoising strategies, such as filling in missing words or reconstructing corrupted input. These methods leverage BERT's ability to understand context in a bidirectional manner while also incorporating generation capabilities.

Building on this, other bidirectional and denoising-based models have been explored for text generation tasks. BART (Lewis (2019)) and MASS (Song (2019)) are two notable models that combine elements of both autoencoding and autoregressive generation. BART, which uses a denoising autoencoder framework, has shown strong performance in various generation tasks by learning to reconstruct corrupted text. MASS, on the other hand, employs a sequence-to-sequence framework where the model is trained to predict masked segments of a sentence. Additionally, T5 (Raffel et al. (2020)) builds upon this approach with a unified framework for a variety of text generation tasks, treating every NLP task as a text-to-text problem.

While BERT and its variants remain largely focused on natural language understanding (NLU), models like BART, MASS, and T5 have extended the capabilities of bidirectional models into the realm of text generation. These models bridge the gap between the traditional strengths of BERT in understanding context and the generation capabilities seen in autoregressive models like GPT.

3 Methodology

In this study, I build upon the experimental work of Berglund et al. (2023) to explore whether bidirectional language models could mitigate the reversal curse. I conduct two main experiments to investigate how models can better generalize to reversed relationships and enhance their ability to infer such relationships. My approach utilizes BART for text generation and BERT for bidirectional inference, with GPT-4-generated synthetic data for context.

3.1 Experiment 1: Utilizing Bart For Text Generation

Berglund et al. (2023) create a dataset made up of documents of the form "$< name >$ is $< description >$" (or the reverse) where the names and descriptions are fictitious. Each description is intended to denote a unique individual. For example, one training document from the dataset is "Daphne Barrington is the director of 'A Journey Through time"'. They use GPT-4 (OpenAI) to generate pairs of names and descriptions.

In this experiment, I use the datasets above to fine-tune BART-BASE model pre-trained for text-generation.

3.2 Experiment 2.1: Modifying the Prompt for Improved Directional Retrieval

Berglund et al. (2023) test GPT-4 on facts about actual celebrities and their parents that have the form "A's parent is B" and "B's child is A". They collect a list of the top 1000 most popular celebrities and query GPT-4 (accessed via the OpenAI API) for their parents. The results are that GPT-4 is able to identify the celebrity's parent 79 % (1513 pairs parent-child relationship) of the time surveyed, but only successful 33% when queried to identify the child from the parent. For instance, it shows that GPT-4 can identify Mary Lee Pfeiffer as Tom Cruise's mother, but can't identify Tom Cruise as Mary Lee Pfeiffer's son.

GPT-4, are trained primarily to focuse more on celebrities and their relationships than their parents. Therefore, When queried with a prompt asking for the reverse relationship, "Name a child of parent," GPT-4 may not have the necessary retrieval mechanisms activated to process such a query effectively. This phenomenon is closely related to findings in prior research by Grosse et al. (2023), which highlights the influence of training examples on language model outputs. Their experiments demonstrate that training examples presented in the order "A precedes B" have a much stronger influence on the model's output than those with the reverse order. This suggests that LLMs trained predominantly on one direction of a relationship fail to generalize well to the reverse direction, which underpins the "reversal curse" hypothesis.

Given the directional bias inherent in the GPT-4 model's training, I hypothesize that modifying the query to reverse the direction of the relationship may activate the necessary latent knowledge within the model to improve its accuracy. Instead of directly asking for the child of a given parent (i.e., "Name a child of {parent}"), I reverse the query direction by asking "Which celebrity has a {parent type} named {parent}". This change aligns more closely with the information or knowledge base that GPT-4 has likely encountered during its training, where the model has more frequently processed more celebrities' relationships than their parents'.

3.3 EXPERIMENT 2.2: BERT-BASED INFERENCE WITH GPT-GENERATED CONTEXT

The BERT Family has been predominantly utilized for extracting contextual features and not for text generation. In this experiment, leveraging the same dataset of child-parent pairs, I employ BERT to deduce the famous child (celebrity) when provided with the parent's name.

To set up this task, I initially harnessed GPT-4 to generate comprehensive details regarding celebrities and their family relationships. The generated response functions as the context for the BERT model.

The pre-trained BERT model employed is bert-large-cased-whole-word-masking-finetuned-squad, which has been specifically fine-tuned for question-answering (QA) tasks. I formulated the problem as a straightforward QA task, where the input to BERT consists of the context generated by GPT-4 (family-related information of {celebrity}) and the question presented to BERT is: "Name a famous child of {parent}".

The context is designed to mimic the knowledge base of a celebrity as GPT-4, aiming to demonstrate BERT's proficiency in identifying the most renowned child (celebrity) within the child-parent network, as opposed to the parent-child relationship. By restricting the potential answers to "famous children", we augment the probability that BERT will retrieve the most prominent child (i.e., celebrity) within the family, rather than any arbitrary child. This was done in an effort to enhance accuracy when the family relationship data is complex.

4 EXPERIMENT & ANALYSIS

4.1 EXPERIMENT 1: UTILIZING BART FOR REVERSING DESCRIPTIONS OF FICTITIOUS CELEBRITIES

Fine-tuning and Model Setup The dataset used in this experiment consists of 3,600 pairs of descriptions and names (or reverse) as training set and 3,00 pairs as validation set.

I use the above dataset to fine-tune the pre-trained BART-BASE model. The aim is to predict the fictitious person given the description, or predict the description given the name.

Evaluation

```
KEYS_WE_CARE_ABOUT = [
    "p2d_reverse_prompts_test",
    "both_prompts_test",
    "p2d_prompts_test",
    "d2p_prompts_test",
    "d2p_reverse_prompts_test",
    "p2d_reverse_prompts_test_randomized",
    "d2p_reverse_prompts_test_randomized",
]
```

Same as the work in Berglund et al. (2023), I use the above 7 different datasets, to evaluate the performance of the fine-tuned model. To achieve this, the input to the model was structured as the description of a celebrity, and the output was the celebrity's name.

The performance of the model is evaluated using two key metrics:

Cosine Similarity: The cosine similarity between the generated completion and the target completion is calculated. This is done using the sentence-transformers library, which encodes the text into dense vector representations. The cosine similarity between the vectors for the completion and target is calculated, which serves as a measure of how similar the generated text is to the target text. A threshold (0.6) is used to classify the completion as "correct" or "incorrect" based on similarity.

Accuracy: The code calculates whether the generated answer matches the target answer exactly. If the cosine similarity is greater than a defined threshold (0.6 in this case), it counts as a correct match.

Results As shown in Figure 1, the fine-tuned BART model performs significantly better on the forward tasks compared to the reverse tasks. In particular, datasets such as p2d_prompts_test_bart and both_prompts_test_bart achieve high accuracy and cosine similarity scores, demonstrating that the model is effective at generating names given descriptions.

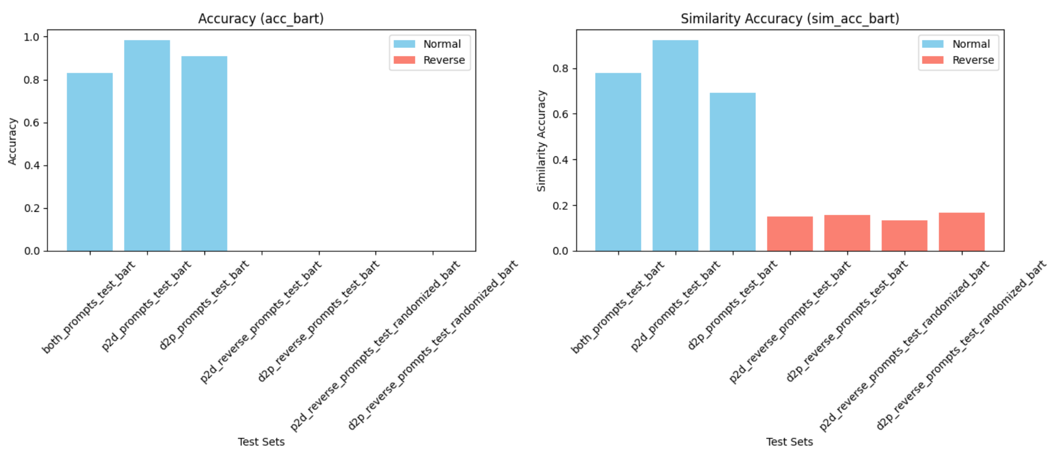

Figure 1: BART's performance on different test set

However, when tested on the reverse tasks (where the input is a name and the output is a description), the performance is much weaker. For example, the accuracy on the p2d_reverse_prompts_test_bart and d2p_reverse_prompts_test_bart datasets is 0.0, with cosine similarities also being very low (around 0.15 and 0.16). The randomized reverse prompts also exhibit similar results, with very low accuracy and similarity scores.

4.2 Experiment 2.1: Modifying the GPT Prompt for REVERSING REAL-WORLD KNOWLEDGE

System Prompt I formulate a system prompt that is primed for recalling facts about various types of celebrities, in order to activate those parts of the network and maximize the chances of retrieval of related knowledge.

```
SYSTEM_PROMPT = f'You are an expert when it comes to any type of
↪ celebrity, including but not limited to actors, singers,
↪ producers and others, and including their family relations.
↪ You answer questions concisely, with only the answer or
↪ {UNKNOWN_STR}'
```

Query Ask "Which celebrity has a {parent type} named {parent}" instead of "Name a child of {parent}"

Results I get 723/1513 correct response while Berglund et al. (2023) only gets 495.

Dividing 723 by 495 tells us that this simple adjustment yields more than 46% better results. The intuition mentioned in above is there, after all, but different approaches have different chances of triggering the activations required to reach it.

4.3 Experiment 2.2: BERT Inference For REVERSING REAL-WORLD KNOWLEDGE

Context Generation GPT-4 is prompted with the following instruction to give the detailed information especially relationships of child(i.e. celebrity):

```
prompt = f"You are a helpful assistant with knowledge of
↪  parent-child relationships. Please provide detailed
↪  information especially name of {child}'s parents, children,
↪  and family relations."
```

Question Formation Once the context is generated, BERT is asked to answer the question: ”Name a famous child of {parent}”.

BERT uses the context generated by GPT-4 and attempts to extract the name of the famous child based on the relationship details provided.

Results The accuracy of BERT in predicting the famous child from the parent's name, using the generated context, reached 70.32%, more than double of the result given by Berglund et al. (2023). This result indicates that BERT is able to effectively reverse the child-parent relationship.

5 Conclusion & Reflection

I explored bidirectional language models to mitigate the reversal curse that large language models (LLMs) face when tasked with reversing relationships between entities. This issue is especially prominent when LLMs, such as GPT-4, struggle to infer relationships in the reverse direction (e.g., identifying a child from a parent). I conducted two main experiments focusing on text generation with BART and bidirectional inference with BERT, using synthetic data generated by GPT-4 to provide context. The results from both experiments reveal valuable insights into the performance of these models in handling reversed relationships and point to promising directions for future research.

Key Findings Experiment 1 showcases that the fine-tuned BART model exhibited high accuracy and cosine similarity for tasks involving generating names given descriptions (forward tasks). However, its performance dropped significantly when tasked with generating descriptions from names (reverse tasks). The limitation in the model's ability to generalize to reversed relationships is out of my expectation. Possible reasons could be the threshold of similarity is not set properly or the scale of the model is not big enough.

Experiment 2.1 suggests that rephrasing the query to align better with the model's training data can activate latent knowledge and improve performance on reversed relationship tasks. It further underscores the importance of query structuring in activating the correct parts of a language model's knowledge base.

Experiment 2.2 uses BERT for bidirectional inference, with context generated by GPT-4, proved to be highly effective. When BERT was provided with detailed family-related information generated by GPT-4, it achieved 70.32% accuracy in identifying the most famous child from the parent's name—more than double the performance reported by **?**. This suggests that BERT, as a bidirectional model, is better equipped to handle reversed relationships than autoregressive models like GPT-4.

Conclusion The results of this study provide strong evidence that bidirectional models, such as BERT, are better suited for tasks involving reversed relationships compared to unidirectional models like GPT-4.

The improvements gained by adjusting GPT-4's prompts also indicate that query modification plays a crucial role in mitigating the reversal curse, suggesting that even autoregressive models can perform better when given the right input structure. These findings open the door for further research into hybrid approaches, combining the strengths of different models (e.g., GPT-4 for context generation and BERT for inference) to more effectively handle reversed relationships.

In summary, this study highlights the potential of bidirectional models like BERT to mitigate the reversal curse, providing a promising direction for future research in natural language processing and the development of models that can more accurately infer and reverse complex relationships.

Reflection Although BERT showed substantial success in reversing relationships, BART—another bidirectional model—did not perform well on reverse tasks. The poor performance of BART in the reverse task highlights that not all bidirectional models may be equally adept at handling reversed relationships, and suggests that further experimentation with other bidirectional models is necessary. For instance, T5 (Text-to-Text Transfer Transformer), which also employs a sequence-to-sequence architecture, could potentially provide better results in this task. Future work could involve fine-tuning T5 or other similar models on the same dataset to assess whether they can further improve performance in reversing relationships.

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